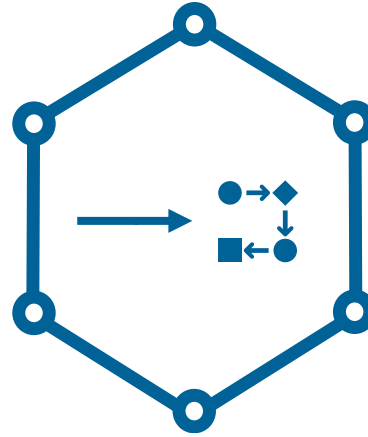




Knowledge Graph Creation Schema Mapping

Bonus

Emanuel Sallinger



STANFORD

Slides courtesy of Vinay K. Chaudhri
University of Stanford

Example Schema Mapping

cookware			
name	type	material	price
c01	skillet	cast iron	50
c02	saucepan	steel	40
c03	skillet	steel	30
c04	saucepan	aluminium	20

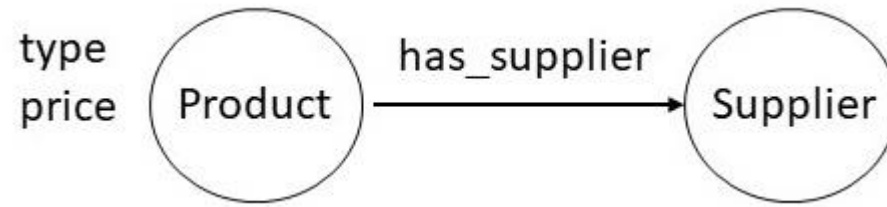
Example Schema Mapping

cookware			
name	type	material	price
c01	skillet	cast iron	50
c02	saucepan	steel	40
c03	skillet	steel	30
c04	saucepan	aluminium	20

kind	
id	value
m01	skillet
m02	skillet
m03	saucepan
m04	saucepan

price	
id	value
m01	60
m02	50
m03	40
m04	20

Example Schema Mapping



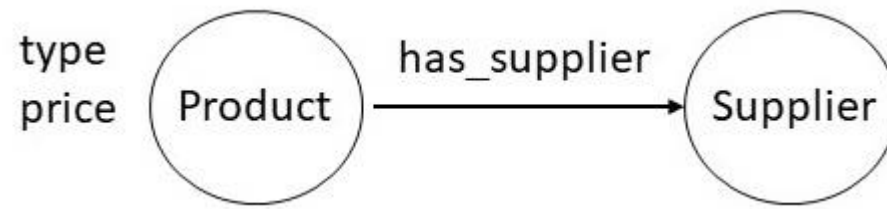
cookware			
name	type	material	price
c01	skillet	cast iron	50
c02	saucepan	steel	40
c03	skillet	steel	30
c04	saucepan	aluminium	20

kind	
id	value
m01	skillet
m02	skillet
m03	saucepan
m04	saucepan

price	
id	value
m01	60
m02	50
m03	40
m04	20

knowledge graph		
subject	predicate	object
c01	type	skillet
c01	price	50
c01	has_supplier	vendor_1
c02	type	saucepan
c02	price	40
c02	has_supplier	vendor_1
c03	type	skillet
c03	price	30
c03	has_supplier	vendor_1
c04	type	saucepan
c04	price	20
c04	has_supplier	vendor_1
m01	type	skillet
m01	price	60
m01	has_supplier	vendor_2
m02	type	skillet
m02	price	50
m02	has_supplier	vendor_2
m03	type	saucepan
m03	price	40
m03	has_supplier	vendor_2
m04	type	saucepan
m04	price	20
m04	has_supplier	vendor_2

Example Schema Mapping



knowledge_graph(ID,type,Type) :- cookware(ID,TYPE,MATERIAL,PRICE)

knowledge_graph(ID,price,PRICE) :- cookware(ID,TYPE,MATERIAL,PRICE)

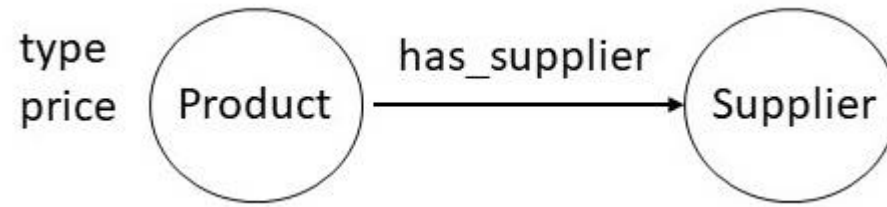
knowledge_graph(ID,has_supplier,vendor_1) :- cookware(ID,TYPE,MATERIAL,PRICE)

cookware			
name	type	material	price
c01	skillet	cast iron	50
c02	saucepan	steel	40
c03	skillet	steel	30
c04	saucepan	aluminium	20

kind	
id	value
m01	skillet
m02	skillet
m03	saucepan
m04	saucepan

price	
id	value
m01	60
m02	50
m03	40
m04	20

Example Schema Mapping



knowledge_graph(ID,type,Type) :- kind(ID,TYPE)

knowledge_graph(ID,price,PRICE) :- price(ID,PRICE)

knowledge_graph(ID,has_supplier,vendor_2) :- kind(ID,TYPE)

cookware			
name	type	material	price
c01	skillet	cast iron	50
c02	saucepan	steel	40
c03	skillet	steel	30
c04	saucepan	aluminium	20

kind	
id	value
m01	skillet
m02	skillet
m03	saucepan
m04	saucepan

price	
id	value
m01	60
m02	50
m03	40
m04	20

Bootstrapping Schema Mapping

- Linguistic Mapping
- Mapping based on instances
- Mapping based on constraints

Bootstrapping Schema Mapping

- Linguistic Techniques
 - Leverage the name
 - Stemming, Synonym, Hypernym
 - Cname and Customer Name
 - Automobile and Vehicle
 - Book and Publication
 - Common substrings/pronunciation
 - Amount Received/Amount Receivable
 - Bell vs Belle
 - Leverage documentation string
 - Extract keywords, and check semantic similarity

Bootstrapping based on Instances

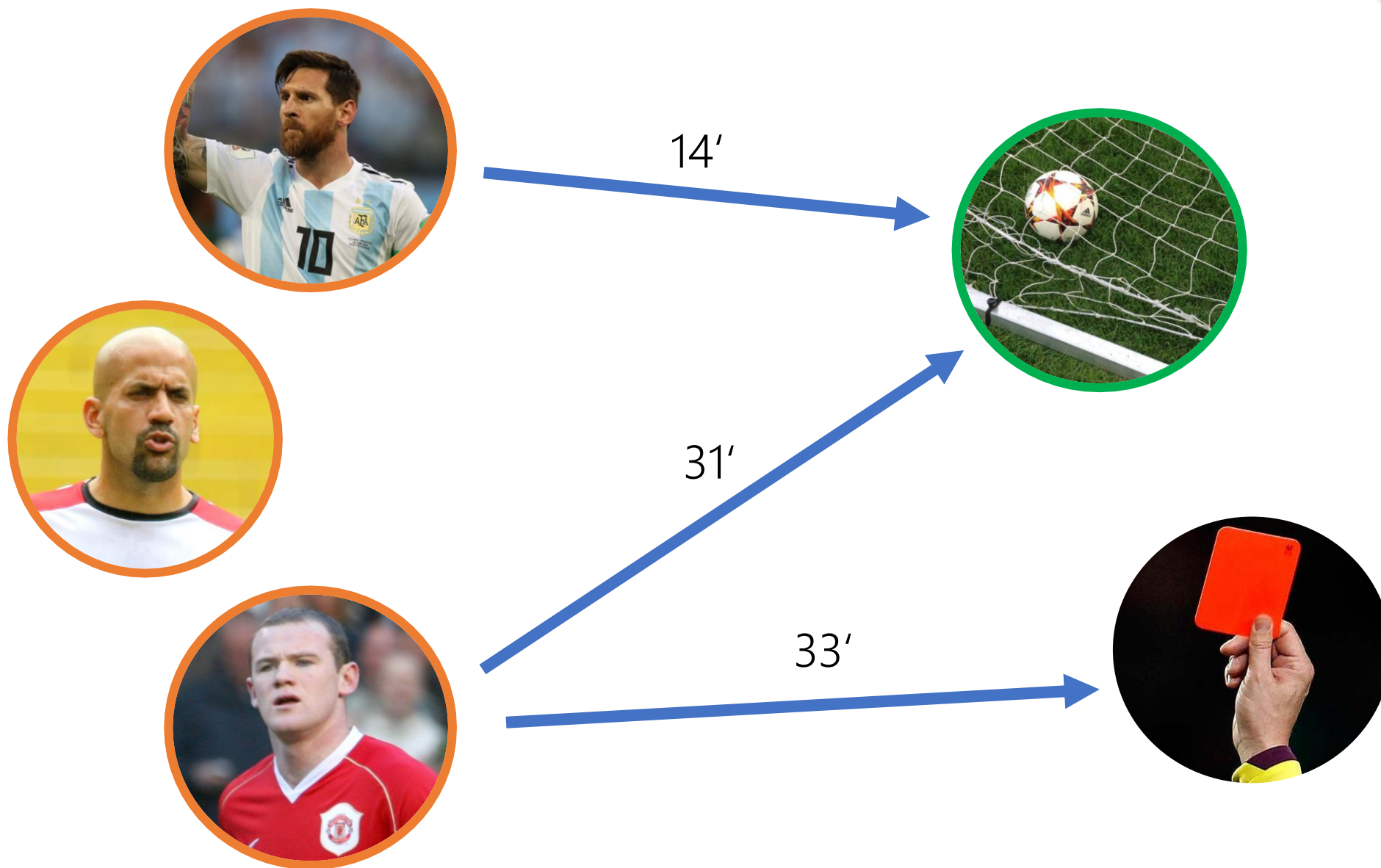
- Examine the data
 - If we can recognize the data contain phone number, zip code, ISBN, SSN, Date that can provide strong guidance for which attributes can match

Bootstrapping based on Constraints

- Leverage the constraints
 - Value range constraints, uniqueness, optionality, cardinality

Bootstrapping Schema Mapping

- Bootstrapping results
 - are inexact
 - need human verification
- Can save some effort
- Lead to a better story



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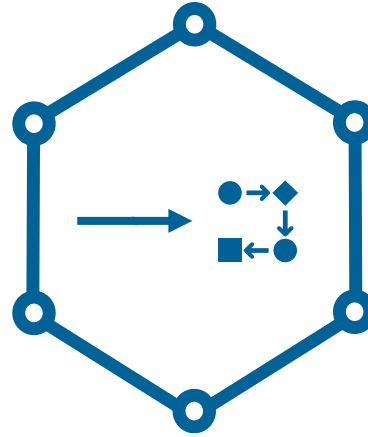
Player		
Team	No.	Name
ARG	10	Messi
ARG	3	Veron
ENG	2	Rooney

Event			
Team	No.	Type	Time
ARG	10	Goal	14'
ENG	2	Goal	31'
ENG	2	Red Card	33'



Report				
Team	No.	Name	Event	Time
ARG	10	Messi	Start	1'
ARG	3	Veron	Start	1'
ENG	2	Rooney	Start	1'

```
report(T,I,N,"start",1) :- player(T,I,N).  
∀t,i,n (Player(t,i,n) → Report(t,i,n,Start,1))
```



Knowledge Graph Creation Schema Mapping

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